

BLM2041 Signals and Systems

Syllabus

The Instructors:

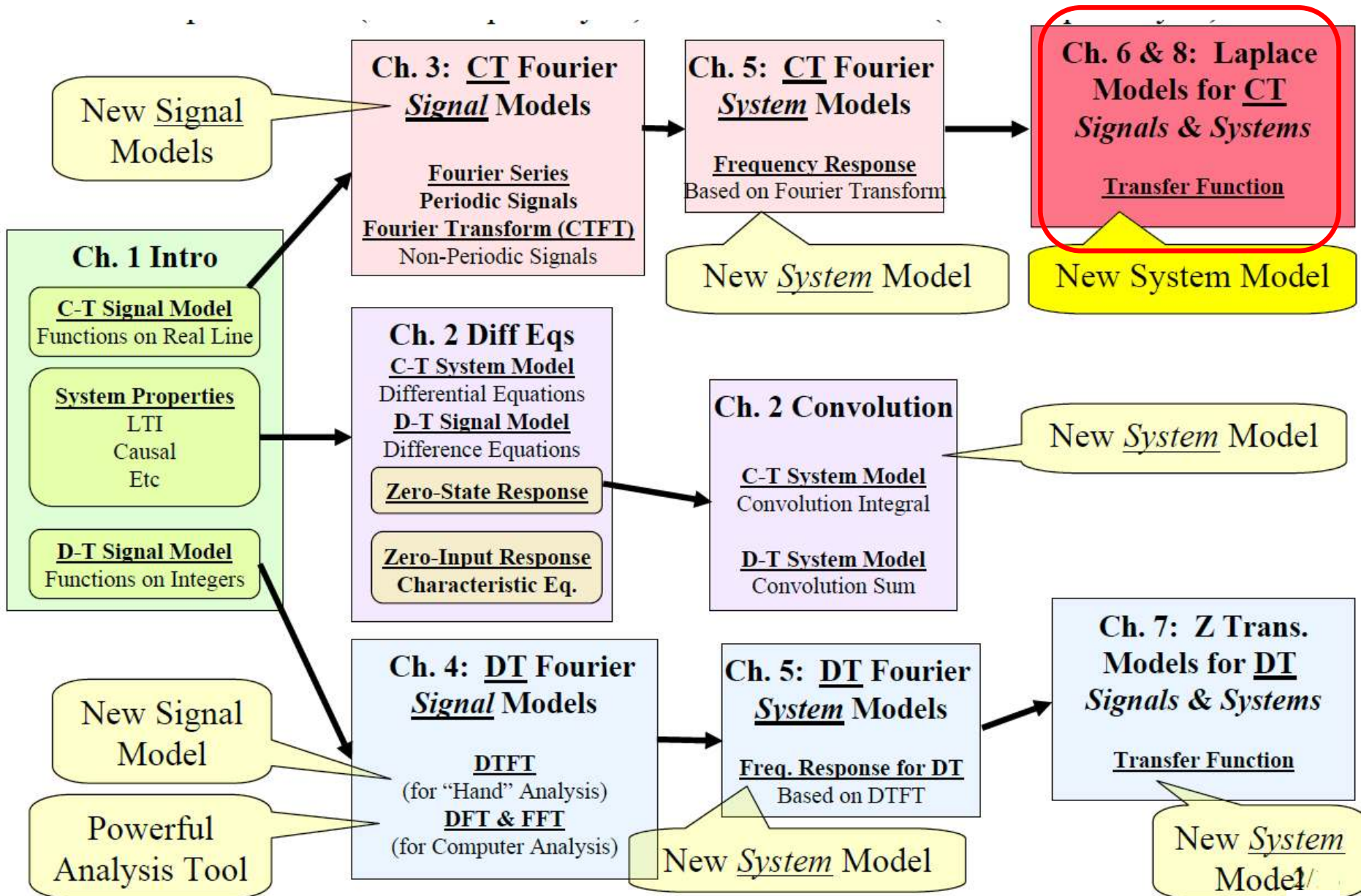
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Where are we now?



What we have seen so far....

- Diff. Equations describe systems
 - Differential Eq. for CT
 - Difference Eq. for DT
- Convolution with the Impulse Response can be used to analyze the system
 - An integral for CT
 - A summation for DT
- Fourier Transform (and Series) describe what frequencies are in a signal
 - CTFT for CT has an integral form
- The Frequency Response of a system gives a multiplicative method of analysis
 - Freq. Response = CTFT of impulse response for CT system

We now look at one “power tool ” for system analysis:

Laplace Transform for CT Systems

Extension of CTFT

What we have seen so far....

Convergence of the Fourier Transform: Finite-Energy Case

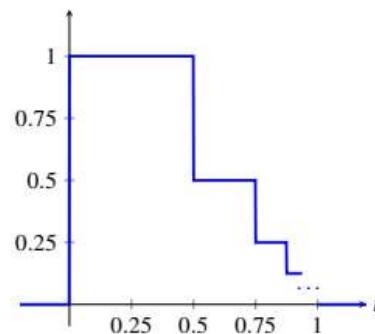
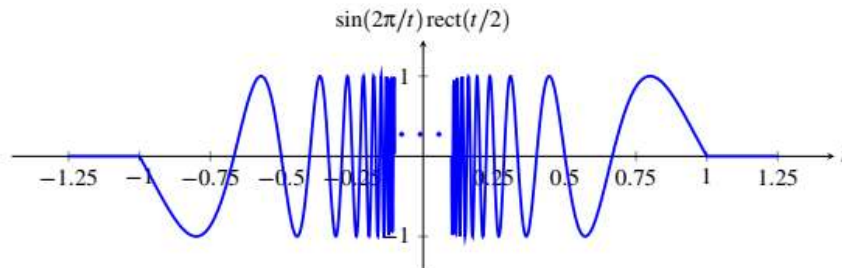
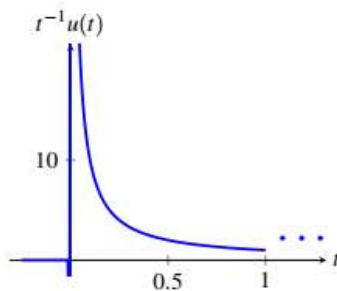
- If a function x is of *finite energy* (i.e., $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$), then its Fourier transform representation converges in the *MSE sense*.
- In other words, if x is of finite energy, then the energy E in the difference function $\tilde{x} - x$ is zero; that is,

$$E = \int_{-\infty}^{\infty} |\tilde{x}(t) - x(t)|^2 dt = 0.$$

- Since, in situations of practical interest, the finite-energy condition in the above theorem is often satisfied, the theorem is frequently applicable.
- It is important to note, however, that the condition $E = 0$ does not necessarily imply $\tilde{x}(t) = x(t)$ for all t .
- Thus, the above convergence result does not provide much useful information regarding the value of $\tilde{x}(t)$ at specific values of t .
- Consequently, the above theorem is typically most useful for simply determining if the Fourier transform representation converges.

Dirichlet Conditions

- The **Dirichlet conditions** for the function x are as follows:
 - 1 the function x is **absolutely integrable** (i.e., $\int_{-\infty}^{\infty} |x(t)| dt < \infty$);
 - 2 on any finite interval, x has a finite number of maxima and minima (i.e., x is of **bounded variation**); and
 - 3 on any finite interval, x has a **finite number of discontinuities** and each discontinuity is itself **finite**.
- Examples of functions violating the Dirichlet conditions are shown below.



Laplace Transform

We've seen that the FT is a useful tool for

- signal analysis (understanding signal structure)
- systems analysis/design

But only if:

1. System is in zero state
2. Impulse response satisfies $\int_{-\infty}^{\infty} |h(t)| dt < \infty$
3. Input satisfies $\int_{-\infty}^{\infty} |x(t)| dt < \infty$

Called
“Absolutely Integrable”



Well... there are a few signals that we can handle with FT that do not satisfy this:
Sinusoids and unit step are two of them

So...frequency response is a tool that can only be used under these three conditions!

The Laplace Transform is a generalization of the CTFT...

it can handle cases when these three conditions are not met.

Generalizing the Fourier Transform

The CTFT expresses a time-domain signal as a linear combination of **complex sinusoids** of the form $e^{j\omega t}$. In the generalization of the CTFT to the Laplace transform, the complex sinusoids become **complex exponentials** of the form e^{st} where s can have any complex value. Replacing the complex sinusoids with complex exponentials leads to this definition of the Laplace transform.

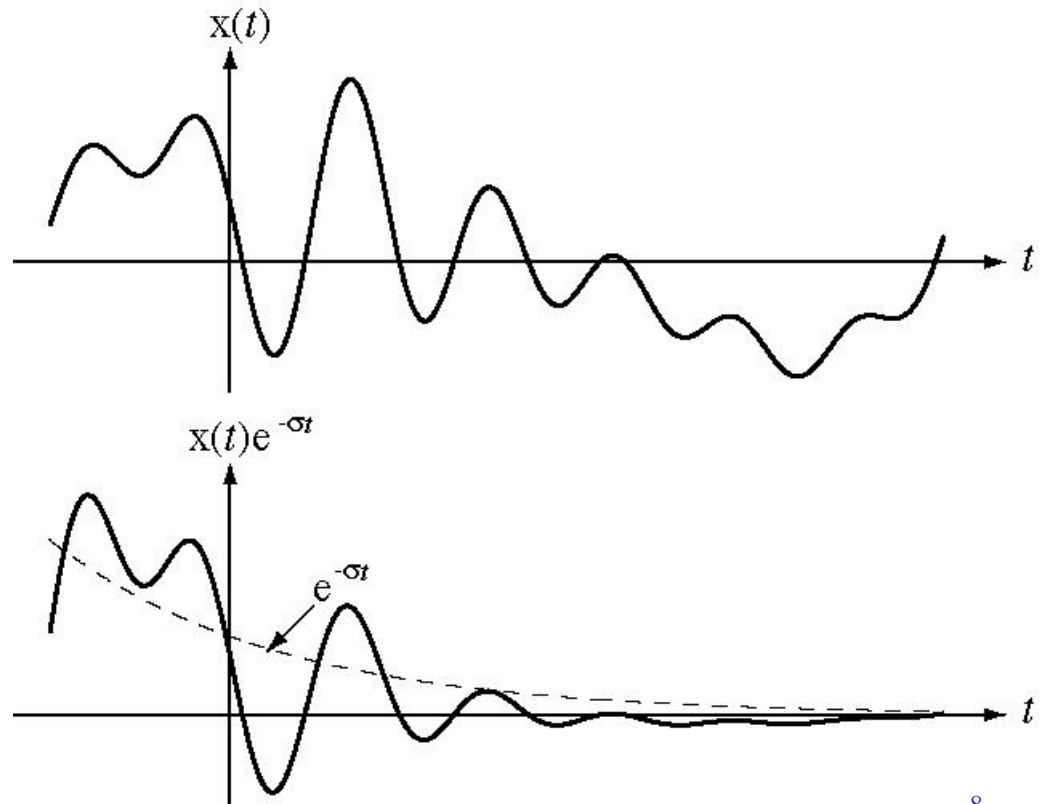
$$\mathcal{L} \left(x(t) \right) = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$
$$x(t) \xleftrightarrow{\mathcal{L}} X(s)$$

Generalizing the Fourier Transform

The variable s is viewed as a generalization of the variable ω of the form $s = \sigma + j\omega$. Then, when σ , the real part of s , is zero, the Laplace transform reduces to the CTFT. Using $s = \sigma + j\omega$ the Laplace transform is

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt \\ &= \mathcal{F} \left[x(t) e^{-\sigma t} \right] \end{aligned}$$

which is the Fourier transform of $x(t) e^{-\sigma t}$



Generalizing the Fourier Transform

The extra factor $e^{-\sigma t}$ is sometimes called a **convergence factor** because, when chosen properly, it makes the integral converge for some signals for which it would not otherwise converge.

For example, strictly speaking, the signal $Au(t)$ does not have a CTFT because the integral does not converge. But if it is multiplied by the convergence factor, and the real part of s is chosen appropriately, the CTFT integral will converge.

$$\int_0^{\infty} Au(t)e^{j\omega t} dt = A \int_0^{\infty} e^{j\omega t} dt \quad \neg \text{ Does not converge}$$

$$\int_0^{\infty} Ae^{-\sigma t} u(t) e^{j\omega t} dt = A \int_0^{\infty} e^{(-\sigma + j\omega)t} dt \quad \neg \text{ Converges (if } \sigma > 0)$$

Laplace Transform

There are two analysis methods that the Laplace Transform enables:

Zero state

LT & Transfer Function

$x(t)$ and $h(t)$ may or may not
be absolutely integrable

So... this just allows us to do the
same thing that the FT does...
but for a larger class of
signals/systems

Non zero-state

LT-based solution of differential
equations

$x(t)$ and $h(t)$ may or may not be
absolutely integrable

This not only admits a larger
class of signals/systems... it also
gives a powerful tool for solving
for both the zero-state AND the
zero-input solutions...

ALL AT ONCE

Laplace Transform Definition

Given a C-T signal $x(t)$ $-\infty < t < \infty$ we've already seen how to use the CTFT:

$$CTFT : X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

Unfortunately the CTFT doesn't "converge" for some signals... the LT mitigates this problem by including decay in the transform:

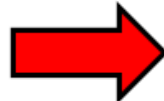
$$e^{-j\omega t} \text{ vs. } e^{-st} = e^{-(\sigma + j\omega)t} = e^{-\sigma t} e^{-j\omega t}$$

Controls decay of integrand



For the Laplace Transform we use: $s = \sigma + j\omega$. So... s is just a complex variable that we almost always view in rectangular form

$$CTFT : X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$



$$LT : X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

Laplace Transform Definition

There are 2 types of LT: 

- Two-sided (bilateral)
- One-sided (unilateral)

Two-Sided LT

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt \quad \text{with } s = \underbrace{\sigma + j\omega}_{\text{complex variable}}$$

One-Sided LT

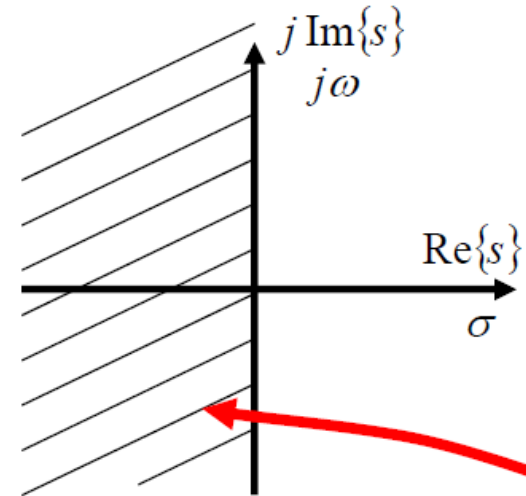
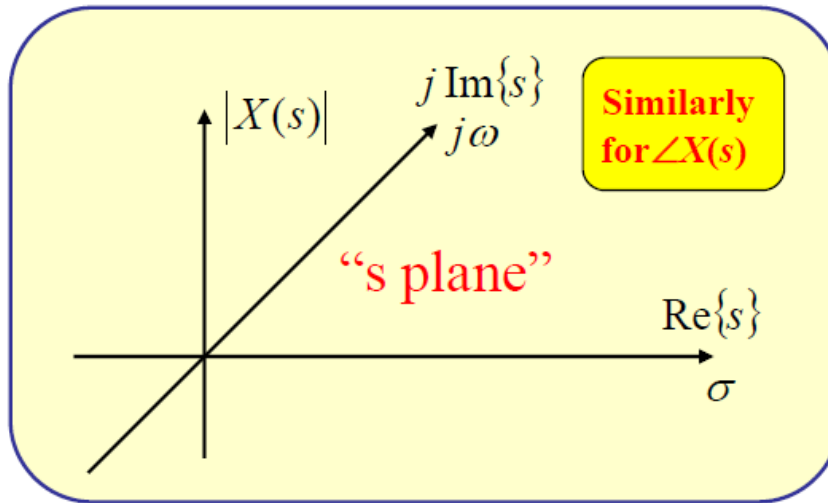
$$X(s) = \int_0^{\infty} x(t)e^{-st} dt \quad \text{with } s = \underbrace{\sigma + j\omega}_{\text{complex variable}}$$

 One-sided LT defined this way $\rightarrow$ even if $x(t) \neq 0, t < 0$

Laplace Transform Definition

Note that $X(s)$ is: $\begin{cases} \text{a complex valued function} \\ \text{of a complex variable } s = \sigma + j\omega \end{cases}$

Must plot on a plane...
the “s-plane”



For the LT we will often need to keep track if s values are in the Left Half Plane (LHP)... so rectangular form of s is convenient. In LHP if $\text{Re}\{s\} < 0$.

Region of Convergence (ROC)

Set of all s values for which the integral in the LT definition converges.

Each signal has its own region of convergence.

Laplace Transform Definition

Example of Finding a LT

Consider the signal $x(t) = e^{-bt}u(t)$ $b \in \mathfrak{R}$

This is a causal signal.

By definition of the LT:

$$X(s) = \int_0^{\infty} e^{-bt} e^{-st} dt = \int_0^{\infty} e^{-(s+b)t} dt$$

This is an easy
integral to do!!

$$X(s) = \frac{-1}{s+b} \left[e^{-(s+b)t} \right]_{t=0}^{t=\infty} = \frac{-1}{s+b} \left[\underbrace{\lim_{t \rightarrow \infty} e^{-(s+b)t}}_{\text{look at this}} - 1 \right]$$

The limit is here by the
definition of the integral

If this limit does not converge... then we say that the integral “does not exist”

So... we need to find out under what conditions this integral exists.

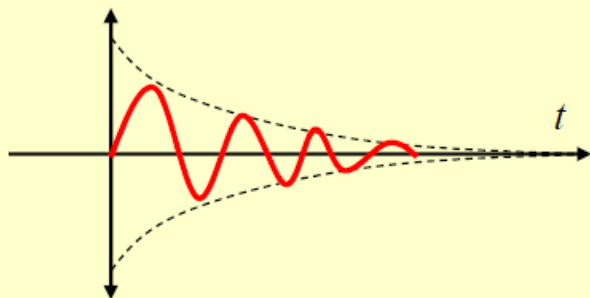
So... let's look at the function inside this limit...

Back when we studied
the FT we had to limit b
to being $b > 0$... with
the LT we don't need to
restrict that!!!

Laplace Transform Definition

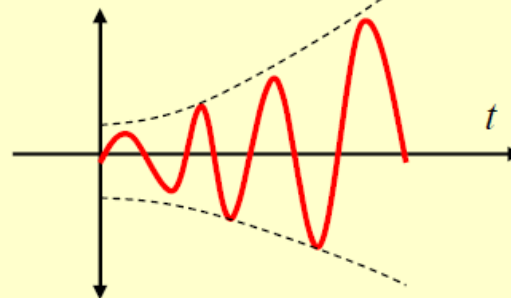
$$e^{-(s+b)t} = e^{-[(\sigma+b)+j\omega]t}$$

$$\text{if } \sigma + b > 0 \Rightarrow \sigma > -b$$



**Has Two
Main
Behaviors**

$$\text{if } \sigma + b < 0 \Rightarrow \sigma < -b$$



Thus, $\lim_{t \rightarrow \infty} e^{-(s+b)t}$ "exists" only for $\sigma > -b$

So, we can't "find" this $X(s)$ for values of s such that $\text{Re}\{s\} \leq -b$

But for s with $\text{Re}\{s\} > -b$ we have no trouble. This set of s is ROC for this transform.

Don't worry too much about ROC... at this level it kind of takes care of itself

So for $x(t) = e^{-bt}u(t)$ We have

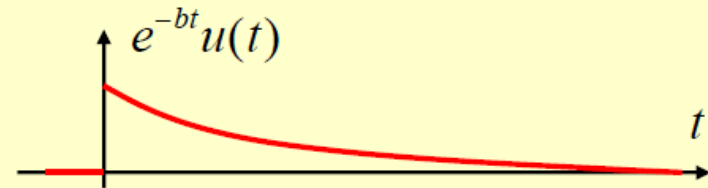
$$X(s) = \frac{1}{s+b} \quad \text{Re}\{s\} > -b$$

This result... and many others... is on the Table of Laplace Transforms

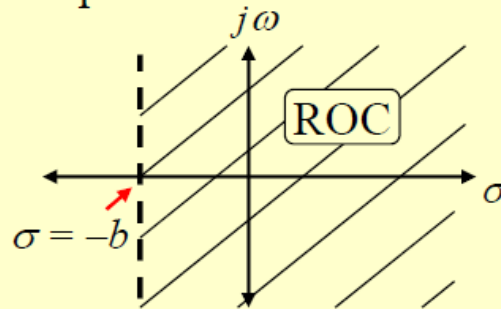
Laplace Transform Definition

If $b > 0$ then $x(t)$ itself decays:

For $b > 0$, $-b$ is negative



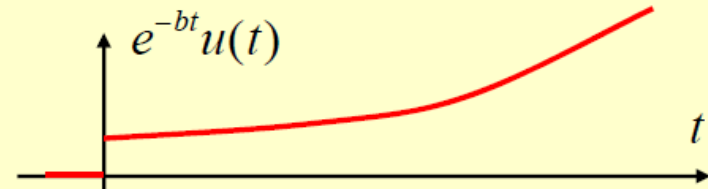
And we have on the s-plane:



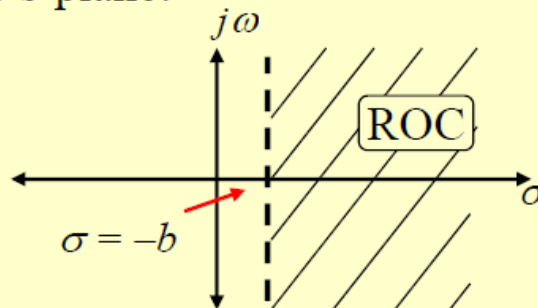
This case can be handled by the FT... and can also be handled by the LT

If $b < 0$ then $x(t)$ itself “explodes”:

For $b < 0$, $-b$ is positive



And we have on the s-plane:



This case can't be handled by the FT... but by restricting our focus to values of s in the ROC, the LT can handle it!!!

Laplace Transform Definition

Connection between CTFT & LT

$$\text{CTFT: } X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

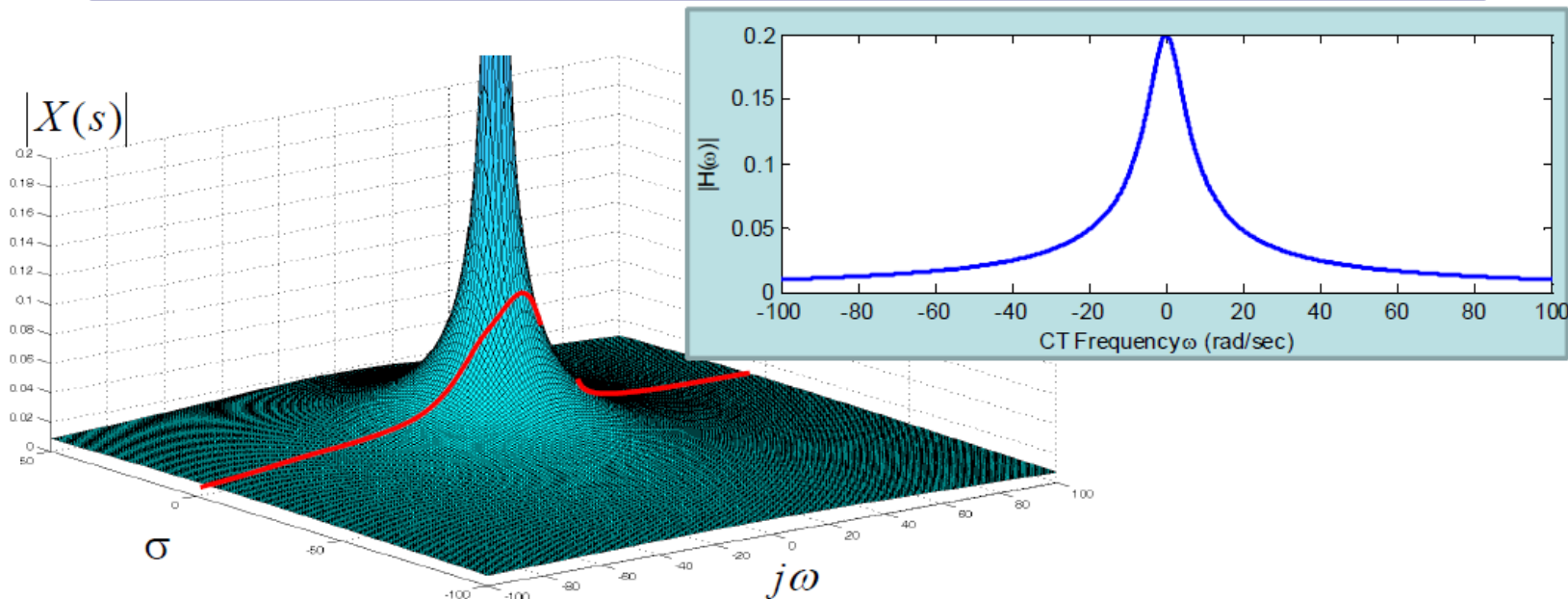
$$\text{LT: } X(\sigma + j\omega) = \int_{-\infty}^{\infty} x(t)e^{-(\sigma + j\omega)t} dt$$

It appears that letting $\sigma = 0$ gives LT = FT...

But this is only true if ROC includes the “ $j\omega$ axis”!!!

If ROC includes “ $j\omega$ axis” Then the FT is “embedded” in the LT

Get the FT by taking the LT and evaluating it only on the $j\omega$ axis...
i.e., take a “slice” of the LT on the $j\omega$ axis (where $\sigma = 0$).

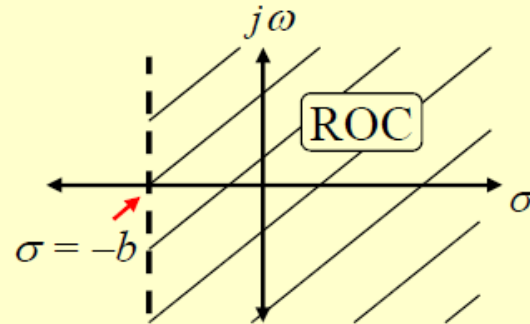


Laplace Transform Definition

Let's Revisit the Example Above

$$x(t) = e^{-bt} u(t) \leftrightarrow X(s) = \frac{1}{s+b} \quad \text{Re}\{s\} > -b$$

If $b > 0$, then ROC includes the “ $j\omega$ axis”:



$$\Rightarrow X(s)|_{s=j\omega} = \left[\frac{1}{s+b} \right]_{s=j\omega} = \underbrace{\left[\frac{1}{j\omega+b} \right]}$$

Same as on
FT table

Region of Convergence

The following two Laplace transform pairs illustrate the importance of the region of convergence.

$$e^{-\alpha t} u(t) \xleftrightarrow{\text{L}} \frac{1}{s + \alpha}, \quad \sigma > -\alpha$$

$$-e^{-\alpha t} u(-t) \xleftrightarrow{\text{L}} \frac{1}{s + \alpha}, \quad \sigma < -\alpha$$

The two time-domain functions are different but the algebraic expressions for their Laplace transforms are the same. Only the ROC's are different.

Region of Convergence

Some of the most common Laplace transform pairs

(There is more extensive table in the book.)

$$\delta(t) \xleftrightarrow{\text{L}} 1, \text{ All } \sigma$$

$$u(t) \xleftrightarrow{\text{L}} 1/s, \sigma > 0$$

$$-u(-t) \xleftrightarrow{\text{L}} 1/s, \sigma < 0$$

$$\text{ramp}(t) = t u(t) \xleftrightarrow{\text{L}} 1/s^2, \sigma > 0$$

$$\text{ramp}(-t) = -t u(-t) \xleftrightarrow{\text{L}} 1/s^2, \sigma < 0$$

$$e^{-\alpha t} u(t) \xleftrightarrow{\text{L}} 1/(s + \alpha), \sigma > -\alpha$$

$$-e^{-\alpha t} u(-t) \xleftrightarrow{\text{L}} 1/(s + \alpha), \sigma < -\alpha$$

$$e^{-\alpha t} \sin(\omega_0 t) u(t) \xleftrightarrow{\text{L}} \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}, \sigma > -\alpha$$

$$-e^{-\alpha t} \sin(\omega_0 t) u(-t) \xleftrightarrow{\text{L}} \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}, \sigma < -\alpha$$

$$e^{-\alpha t} \cos(\omega_0 t) u(t) \xleftrightarrow{\text{L}} \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}, \sigma > -\alpha$$

$$-e^{-\alpha t} \cos(\omega_0 t) u(-t) \xleftrightarrow{\text{L}} \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}, \sigma < -\alpha$$

The Transfer Function

Let $x(t)$ be the excitation and let $y(t)$ be the response of a system with impulse response $h(t)$. The Laplace transform of $y(t)$ is

$$Y(s) = \int_{-\infty}^{\infty} y(t) e^{-st} dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau \right] e^{-st} dt$$

$$Y(s) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau \right) e^{-st} dt$$

$$Y(s) = \int_{-\infty}^{\infty} h(\tau) d\tau \int_{-\infty}^{\infty} x(t-\tau) e^{-st} dt$$

The Transfer Function

Let $x(t) = u(t)$ and let $h(t) = e^{-4t} u(t)$. Find $y(t)$.

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} u(\tau) e^{-4(t - \tau)} u(t - \tau) d\tau$$

$$y(t) = \begin{cases} \int_0^t e^{-4(t - \tau)} d\tau = e^{-4t} \int_0^t e^{4\tau} d\tau = e^{-4t} \frac{e^{4\tau}}{4} \Big|_0^t = \frac{1 - e^{-4t}}{4}, & t > 0 \\ 0, & t < 0 \end{cases}$$

$$y(t) = (1/4)(1 - e^{-4t})u(t)$$

$$X(s) = 1/s, H(s) = \frac{1}{s + 4} \Rightarrow Y(s) = \frac{1}{s} \times \frac{1}{s + 4} = \frac{1/4}{s} - \frac{1/4}{s + 4}$$

$$x(t) = (1/4)(1 - e^{-4t})u(t)$$

Laplace Transform Example

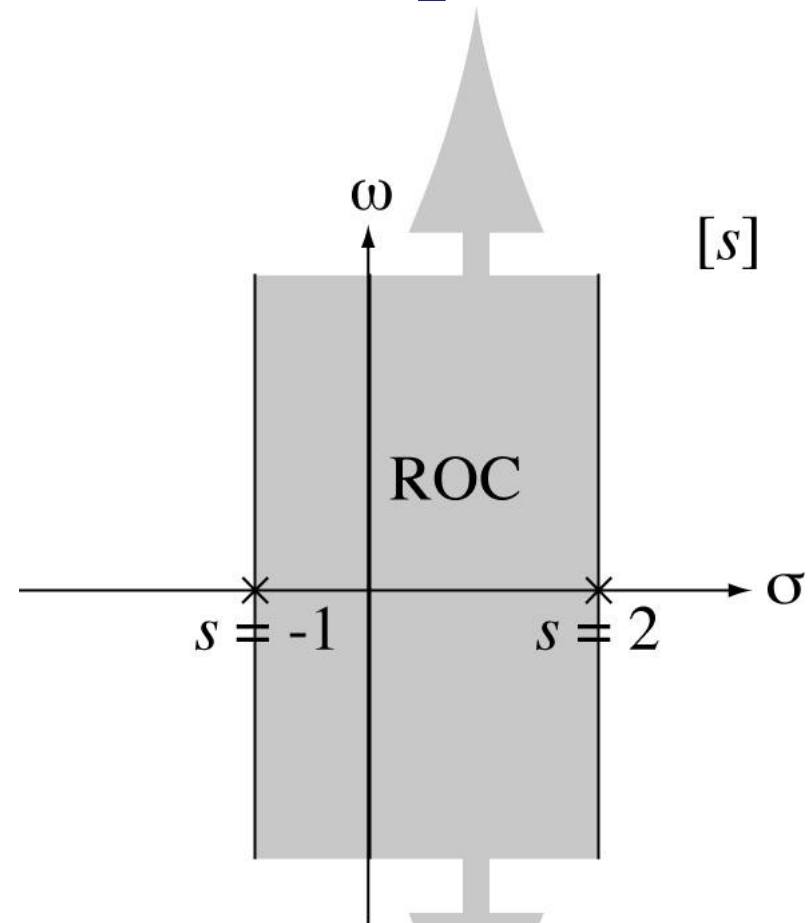
Find the Laplace transform of

$$x(t) = e^{-t} u(t) + e^{2t} u(-t)$$

$$e^{-t} u(t) \xleftrightarrow{\text{L}} \frac{1}{s+1}, \quad \sigma > -1$$

$$e^{2t} u(-t) \xleftrightarrow{\text{L}} -\frac{1}{s-2}, \quad \sigma < 2$$

$$e^{-t} u(t) + e^{2t} u(-t) \xleftrightarrow{\text{L}} \frac{1}{s+1} - \frac{1}{s-2}, \quad -1 < \sigma < 2$$



Laplace Transform Example

Find the inverse Laplace transform of

$$X(s) = \frac{4}{s+3} - \frac{10}{s-6}, \quad 3 < \sigma < 6$$

The ROC tells us that $\frac{4}{s+3}$ must inverse transform into a

right-sided signal and that $\frac{10}{s-6}$ must inverse transform into

a left-sided signal.

$$x(t) = 4e^{-3t} u(t) + 10e^{6t} u(-t)$$

Laplace Transform Example

Find the inverse Laplace transform of

$$X(s) = \frac{4}{s+3} - \frac{10}{s-6}, \quad \text{Re}(s) > 6$$

The ROC tells us that both terms must inverse transform into a right-sided signal.

$$x(t) = 4e^{-3t} u(t) - 10e^{6t} u(t)$$

Laplace Transform Example

Find the inverse Laplace transform of

$$X(s) = \frac{4}{s+3} - \frac{10}{s-6}, \quad \sigma < -3$$

The ROC tells us that both terms must inverse transform into a left-sided signal.

$$x(t) = 4e^{-3t} u(-t) + 10e^{6t} u(-t)$$

MATLAB System Objects

A MATLAB system object is a special kind of variable in MATLAB that contains all the information about a system.

It can be created with the `tf` command whose syntax is

$$\text{sys} = \text{tf}(\text{num}, \text{den})$$

where `num` is a vector of numerator coefficients of powers of s , `den` is a vector of denominator coefficients of powers of s , both in descending order and `sys` is the system object.

MATLAB System Objects

For example, the transfer function

$$H_1(s) = \frac{s^2 + 4}{s^5 + 4s^4 + 7s^3 + 15s^2 + 31s + 75}$$

can be created by the commands

```
»num = [1 0 4] ; den = [1 4 7 15 31 75] ;
```

```
»H1 = tf(num,den) ;
```

```
»H1
```

Transfer function:

$$s^2 + 4$$

$$s^5 + 4 s^4 + 7 s^3 + 15 s^2 + 31 s + 75$$

Properties of Bilateral Laplace Transform

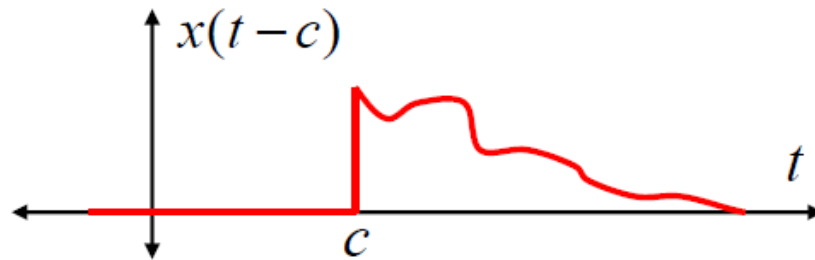
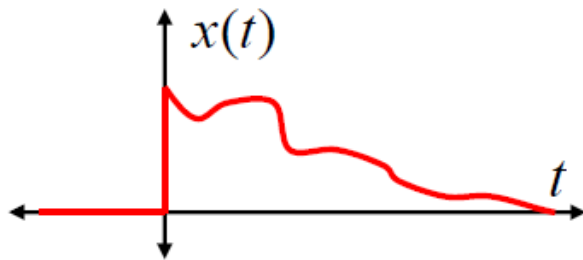
Because of the connection between FT & LT we expect these to be similar to the FT properties we already know!

Linearity: $ax(t) + by(t) \leftrightarrow aX(s) + bY(s)$

There are several other properties... they are listed on the Table of Laplace Transform Properties.

Time Shift :

Figures here for show causal signal (but result is general case)



$$x(t-c) \leftrightarrow e^{-cs} X(s)$$

Compare to time shift for FT: $e^{-j\omega c}$ vs. e^{-cs}

Recall: $s = \sigma + j\omega$

Properties of Bilateral Laplace Transform

Time Differentiation:

$$\dot{x}(t) \leftrightarrow sX(s)$$

Integration:

$$\int_{-\infty}^t x(\lambda) d\lambda \leftrightarrow \frac{1}{s} X(s)$$

These two properties have a nice “opposite” relationship:

Note:	Differentiation	$\Rightarrow$	Multiply by s
	Integration	$\Rightarrow$	Divide by s

“opposites”

“opposites”

These two properties are crucial for linking the LT to the solution of Diff. Eq.

They are also crucial for thinking about “system block diagrams”

Properties of Bilateral Laplace Transform

Time Scaling:

Compare to FT

$$x(at) \leftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right) \quad a > 0$$

Note: $a < 0$ makes $x(at)$ non-causal
So we limit to $a > 0$

Multiply by t^n :

$$t^n x(t) \leftrightarrow (-1)^N \frac{d^N}{ds^N} X(s)$$

Multiply by Exponential:

$$e^{at} x(t) \leftrightarrow X(s - a)$$

with a real or complex

Shift in s-plane

Multiply by sinusoid:

$$x(t) \sin(\omega_0 t) \leftrightarrow \frac{j}{2} [X(s + j\omega_0) - X(s - j\omega_0)]$$

$$x(t) \cos(\omega_0 t) \leftrightarrow \frac{j}{2} [X(s + j\omega_0) + X(s - j\omega_0)]$$

Properties of Bilateral Laplace Transform

Convolution: $x(t) * h(t) \leftrightarrow X(s)H(s)$

Same as for FT!

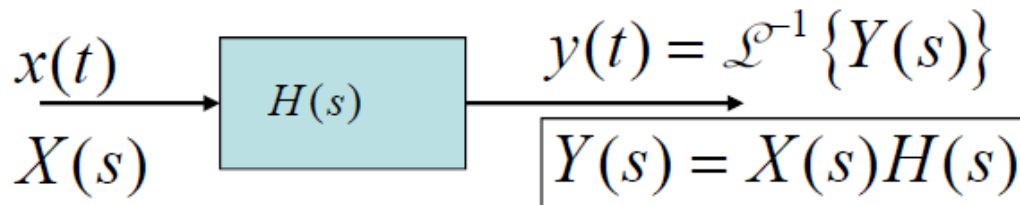
Recall: If $h(t)$ is system impulse response
then $H(\omega)$ is system Frequency Response

We'll see that $H(s)$ is system "Transfer Function"



"Transfer Function"
is a
generalization of
"Frequency Response"

So we have:



Inverse Laplace Transform

Like the FT...once you know $X(s)$ you can use the inverse LT to get $x(t)$

The definition of the inverse LT is:

$$x(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} X(s) e^{st} ds$$

with c chosen such that $s = c + j\omega$ is in ROC

This is a “complex line integral” in complex s-plane...

HARD TO DO!!

But...if
$$X(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_1 s + b_0}{a_N s^N + a_{N-1} s^{N-1} + \dots + a_1 s + a_0}$$

Ratio of polynomials in s
“Rational Function”

Then its easy to find $x(t)$ using partial fraction expansion and a table of LT pairs

Ex. #1: No Direct Terms, No Repeated Roots, No Complex Roots

$$Y(s) = \frac{3s - 1}{s^2 + 3s + 2}$$

If the highest power in the numerator is less than the highest power in the denominator then there will be no direct terms.

By using the quadratic formula to find the roots of the denominator we can verify that there are no repeated or complex roots.

The roots are: $s = -2$ and $s = -1$... so we can write: $Y(s) = \frac{3s - 1}{(s + 1)(s + 2)}$

If there are no repeated roots and no direct terms we can always write it as

$$Y(s) = \frac{r_1}{(s + 1)} + \frac{r_2}{(s + 2)}$$

The numbers r_1 and r_2 are called the “residues”... we need to find them!

Now we exploit what we know:

$$\frac{3s-1}{(s+1)(s+2)} = \frac{r_1}{(s+1)} + \frac{r_2}{(s+2)}$$

Multiply each side by $(s+1)$ gives:

$$\frac{(3s-1)(s+1)}{(s+1)(s+2)} = \frac{r_1(s+1)}{(s+1)} + \frac{r_2(s+1)}{(s+2)}$$

Canceling $(s+1)$ where we can gives:

$$\frac{(3s-1)}{(s+2)} = r_1 + \frac{r_2(s+1)}{(s+2)}$$

Setting $s = -1$ gives:

$$\frac{(-3-1)}{(-1+2)} = r_1 \Rightarrow r_1 = -4$$

All of this is
summarized
by this

$$r_1 = Y(s)(s+1)\Big|_{s=-1}$$

Partial Fractions

When trying to find the inverse Laplace transform it is helpful to be able to break a complicated ratio of two polynomials into forms that are on the Laplace Transform table.

An Example:

$$\mathcal{L}^{-1}\left\{\left[\frac{1/RC}{(s+1/RC)s}\right]\right\} = \mathcal{L}^{-1}\left\{\left[\frac{1}{s} - \frac{1}{s+1/RC}\right]\right\}$$

Factor LT into simpler forms

$$= \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+1/RC}\right\}$$

Linearity of LT

$$= u(t) - e^{-(t/RC)}u(t)$$

Here the factoring could be done “by inspection”... in general, this factoring can be done using the method of “partial fraction expansion” (PFE)

Partial Fractions

Motivation... Going the “Other Way”

Suppose we had this:

$$Y(s) = \frac{2}{(s+3)} + \frac{4}{(s+5)} + 2$$

This has poles at
 $s = -3$ and $s = -5$

A term like this
is called a
“Direct Term”

We could find the common denominator and combine:

$$Y(s) = \frac{2(s+5)}{(s+3)(s+5)} + \frac{4(s+3)}{(s+3)(s+5)} + 2 \frac{(s+3)(s+5)}{(s+3)(s+5)}$$

$$= \frac{2s^2 + 22s + 47}{s^2 + 8s + 15}$$

We want to go from
here... to here...

Similarly... we find the other residue using:

$$r_2 = Y(s)(s+2)\big|_{s=-2} = 7$$

Then we have:

$$Y(s) = \frac{-4}{(s+1)} + \frac{7}{(s+2)}$$

Each of these terms is on the LT Table, so we get

$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\left\{\frac{-4}{(s+1)}\right\} + \mathcal{L}^{-1}\left\{\frac{7}{(s+2)}\right\} \\ &= -4e^{-t}u(t) + 7e^{-2t}u(t) \end{aligned}$$

Partial-Fraction Expansion

The inverse Laplace transform can always be found (in principle at least) by using the inversion integral. But that is rare in engineering practice. The most common type of Laplace-transform expression is a ratio of polynomials in s ,

$$G(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_1 s + b_0}{s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0}$$

The denominator can be factored, putting it into the form,

$$G(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_1 s + b_0}{(s - p_1)(s - p_2) \cdots (s - p_N)}$$

Partial-Fraction Expansion

For now, assume that there are no repeated poles and that $N > M$, making the fraction **proper** in s . Then it is possible to write the expression in the **partial fraction** form,

$$G(s) = \frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} + \dots + \frac{K_N}{s - p_N}$$

where

$$\frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_1 s + b_0}{(s - p_1)(s - p_2) \dots (s - p_N)} = \frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} + \dots + \frac{K_N}{s - p_N}$$

The K 's can be found by any convenient method.

Partial-Fraction Expansion

Multiply both sides by $s - p_1$

$$(s - p_1) \frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_1 s + b_0}{(s - p_1)(s - p_2) \dots (s - p_N)} = \left[K_1 + (s - p_1) \frac{K_2}{s - p_2} + \dots \right. \\ \left. + (s - p_1) \frac{K_N}{s - p_N} \right]$$

$$K_1 = \frac{b_M p_1^M + b_{M-1} p_1^{M-1} + \dots + b_1 p_1 + b_0}{(p_1 - p_2) \dots (p_1 - p_N)}$$

All the K 's can be found by the same method and the inverse Laplace transform is then found by table look-up.

Partial-Fraction Expansion

$$H(s) = \frac{10s}{(s+4)(s+9)} = \frac{K_1}{s+4} + \frac{K_2}{s+9} \quad , \quad > 4$$

$$K_1 = \left[\cancel{(s+4)} \frac{10s}{\cancel{(s+4)}(s+9)} \right]_{s=4} = \left[\frac{10s}{s+9} \right]_{s=4} = \frac{40}{5} = 8$$

$$K_2 = \left[\cancel{(s+9)} \frac{10s}{(s+4)\cancel{(s+9)}} \right]_{s=9} = \left[\frac{10s}{s+4} \right]_{s=9} = \frac{90}{5} = 18$$

$$H(s) = \frac{8}{s+4} + \frac{18}{s+9} = \frac{8s + 72 + 18s + 72}{(s+4)(s+9)} = \frac{10s}{(s+4)(s+9)} \quad . \quad \text{Check.}$$

↓ ↓ ↓ ↓ ↓

$$h(t) = (8e^{-4t} + 18e^{-9t})u(t)$$

Partial-Fraction Expansion

If the expression has a repeated pole of the form,

$$G(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_1 s + b_0}{(s - p_1)^2 (s - p_3) \dots (s - p_N)}$$

the partial fraction expansion is of the form,

$$G(s) = \frac{K_{12}}{(s - p_1)^2} + \frac{K_{11}}{s - p_1} + \frac{K_3}{s - p_3} + \dots + \frac{K_N}{s - p_N}$$

and K_{12} can be found using the same method as before.

But K_{11} cannot be found using the same method.

Partial-Fraction Expansion

Instead K_{11} can be found by using the more general formula

$$K_{qk} = \frac{1}{(m-k)!} \frac{d^{m-k}}{ds^{m-k}} \left[(s - p_q)^m H(s) \right]_{s \rightarrow p_q}, \quad k = 1, 2, \dots, m$$

where m is the order of the q th pole, which applies to repeated poles of any order.

If the expression is not a proper fraction in s the partial-fraction method will not work. But it is always possible to **synthetically divide** the numerator by the denominator until the remainder is a proper fraction and then apply partial-fraction expansion.

Partial-Fraction Expansion

$$H(s) = \frac{10s}{(s+4)^2(s+9)} = \frac{K_{12}}{(s+4)^2} + \frac{K_{11}}{s+4} + \frac{K_2}{s+9}, \quad s > -4$$

↑

Repeated Pole

$$K_{12} = \left[\cancel{(s+4)^2} \frac{10s}{\cancel{(s+4)^2} (s+9)} \right]_{s=-4} = \frac{40}{5} = 8$$

Using

$$K_{qk} = \frac{1}{(m-k)!} \frac{d^{m-k}}{ds^{m-k}} \left[(s-p_q)^m H(s) \right]_{s \rightarrow p_q}, \quad k=1,2,\dots,m$$

$$K_{11} = \frac{1}{(2-1)!} \frac{d^{2-1}}{ds^{2-1}} \left[(s+4)^2 H(s) \right]_{s \rightarrow -4} = \frac{d}{ds} \left[\frac{10s}{s+9} \right]_{s \rightarrow -4}$$

Partial-Fraction Expansion

$$K_{11} = \left[\frac{(s+9)10 - 10s}{(s+9)^2} \right]_{s=-4} = \frac{18}{5}$$

$$K_2 = \frac{18}{5} \Rightarrow H(s) = \frac{8}{(s+4)^2} + \frac{18/5}{s+4} + \frac{18/5}{s+9}, \quad s > -4$$

$$H(s) = \frac{8s^2 + 72 + \frac{18}{5}(s^2 + 13s + 36) + \frac{18}{5}(s^2 + 8s + 16)}{(s+4)^2(s+9)}, \quad s > -4$$

$$H(s) = \frac{10s}{(s+4)^2(s+9)}, \quad s > -4$$

$$h(t) = \left(8te^{-4t} + \frac{18}{5}e^{-4t} - \frac{18}{5}e^{-9t} \right) u(t)$$

Partial-Fraction Expansion

$$H(s) = \frac{10s^2}{(s+4)(s+9)}, \quad > 4 \leftarrow \text{Improper in } s$$

$$H(s) = \frac{10s^2}{s^2 + 13s + 36}, \quad > 4$$

$$\text{Synthetic Division} \rightarrow s^2 + 13s + 36 \overline{) 10s^2} \quad \begin{array}{r} 10 \\ 10s^2 + 130s + 360 \\ \hline 130s \quad 360 \end{array}$$

$$H(s) = 10 + \frac{130s + 360}{(s+4)(s+9)} = 10 \left[\frac{32}{s+4} + \frac{162}{s+9} \right], \quad > 4$$

$$h(t) = 10 \delta(t) + [162e^{-9t} - 32e^{-4t}]u(t)$$

Partial-Fraction Expansion

Complex root situation

$$\mathcal{L}^{-1} \left\{ \frac{s+1}{s^2+2s+17} \right\} \rightarrow s_1 = -1 + 4i, s_2 = -1 - 4i$$

$$\mathcal{L}^{-1} \left\{ \frac{s+1}{(s+1)^2+16} \right\} = \mathcal{L}^{-1} \left\{ \frac{s+1}{(s+1)^2+4^2} \right\} = \mathbf{e^{-t} \cdot \cos(4t)}$$

$$\mathcal{L}^{-1} \left\{ \frac{s-3}{s^2-6s+34} \right\} \rightarrow s_1 = 3 + 5i, s_2 = 3 - 5i$$

$$\mathcal{L}^{-1} \left\{ \frac{s-3}{(s-3)^2+25} \right\} = \mathcal{L}^{-1} \left\{ \frac{s-3}{(s-3)^2+5^2} \right\} = \mathbf{e^{3t} \cdot \cos(5t)}$$

$$\mathcal{L}^{-1} \left\{ \frac{2}{s^2-6s+13} \right\} \rightarrow s_1 = 3 + 2i, s_2 = 3 - 2i$$

$$\mathcal{L}^{-1} \left\{ \frac{2}{(s-3)^2+4} \right\} = \mathcal{L}^{-1} \left\{ \frac{2}{(s-3)^2+2^2} \right\} = \mathbf{e^{3t} \cdot \sin(2t)}$$

Inverse Laplace Transform Example

Method 3

A can be found as before to be $3/2$. Letting $s = 0$, the identity becomes $0 = \frac{3/2}{3} + \frac{C}{5}$ and $C = 5/2$. Then, letting $s = 1$, and solving we get $B = 3/2$. Now

$$G(s) = \frac{3/2}{s-3} + \frac{(3/2)s + 5/2}{s^2 - 4s + 5}, \quad < 2$$

or

$$G(s) = \frac{3/2}{s-3} - \frac{3}{2} \frac{s-2}{s^2 - 4s + 5}, \quad < 2$$

This is the same as a result in Method 2 and the rest of the solution is also the same. The advantage of this method is that all the numbers are real.

Method 2

$$G(s) = \frac{3/2}{s-3} - \frac{3}{2} \left[\frac{s-2}{(s-2)^2 + 1} + \frac{1/3}{(s-2)^2 + 1} \right], \quad < 2$$

This can now be directly inverse Laplace transformed into

$$g(t) = (3/2) \left\{ e^{2t} [\cos(t) + (1/3)\sin(t)] - e^{3t} \right\} u(t)$$

which is the same as the previous result.

Using LT to Solve Differential Equations

$$y_{total}(t) = \underbrace{y_{zs}(t)} + \underbrace{y_{zi}(t)}$$

We've seen how to find this using:
“convolution w/ impulse response”

or using

“multiplication w/ frequency response”

We've seen how to find this
using the characteristic
equation, its roots, and the so-
called “characteristic modes”

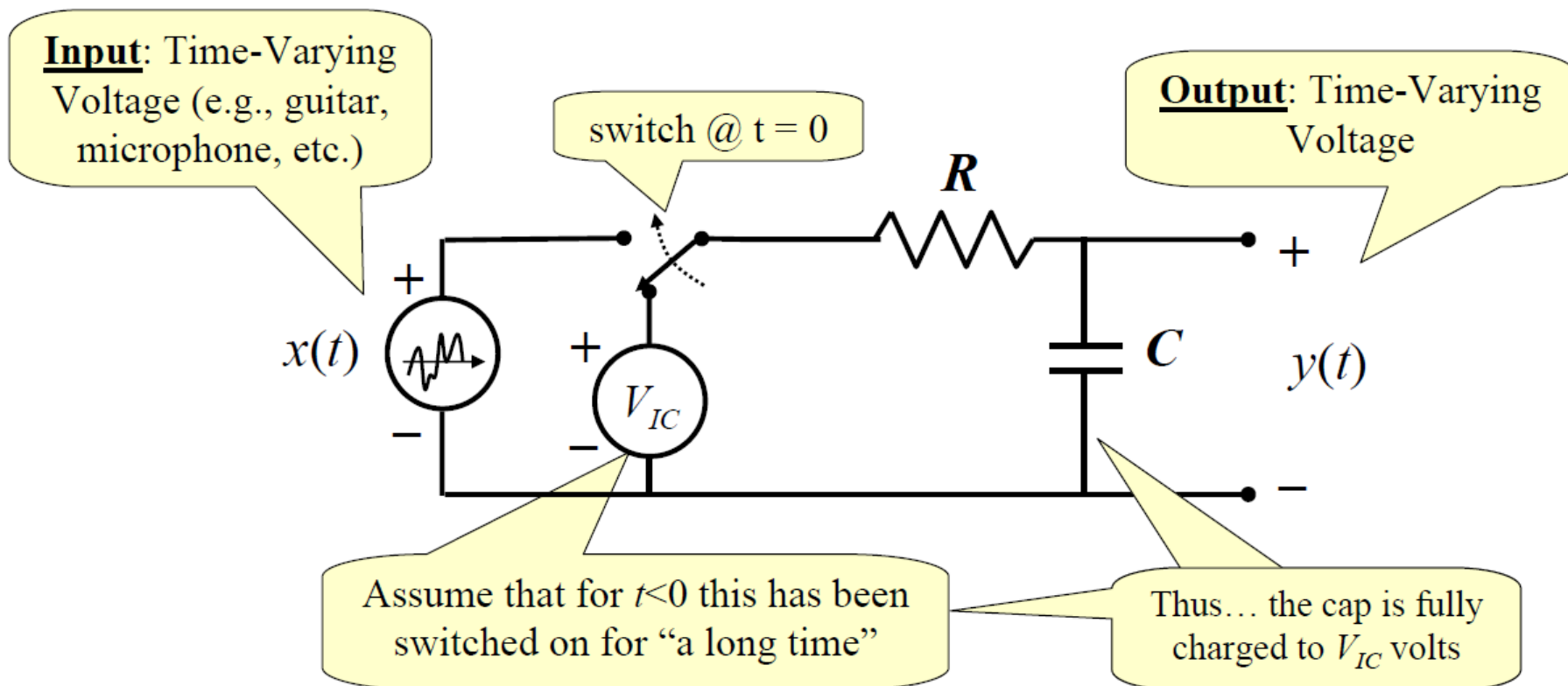
**Here we'll see how to get $y_{total}(t)$ using LT...
... get both parts with one tool!!!**

First-order case: Let's see this for a **1st-order Diff. Eq.** with a **causal input** and a **non-zero initial condition** just before the causal input is applied.

The 1st-order Diff. Eq. describes: a simple RC or RL circuit.

The causal input means: we switch on some input at time $t = 0$.

The initial condition means: just before we switch on the input the capacitor has a specified voltage on it (i.e., it holds some charge).



This circuit is then described by this Diff. Eq.:

$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$$

Cap voltage... just
before $x(t)$ “turns on”

With IC $y(0^-) = V_{IC}$

$$x(t) = 0, \quad t < 0$$

For this ex. we'll solve the general 1st-order Diff. Eq.:

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

Now the **key steps** in using the LT are:

- take the LT of both sides of the Differential Equation...
- use the LT properties where appropriate...
- solve the resulting Algebraic Equation for $Y(s)$
- find the inverse LT of the resulting $Y(s)$

Laplace Transform:

Differential Equation...

turns into an...

Algebraic Equation

Hard to solve

Easy to solve

We now apply these steps to the 1st-order Diff. Eq.:

$$\mathcal{L}\left\{\frac{dy(t)}{dt} + ay(t)\right\} = \mathcal{L}\{bx(t)\}$$

Apply LT to both sides

$$\mathcal{L}\left\{\frac{dy(t)}{dt}\right\} + a\mathcal{L}\{y(t)\} = b\mathcal{L}\{x(t)\}$$

Use Linearity of LT

$$[sY(s) - y(0^-)] + aY(s) = bX(s)$$

Use Property for LT of Derivative... accounting for the IC

$$Y(s) = \frac{y(0^-)}{s+a} + \frac{b}{s+a}X(s)$$

Solve algebraic equation for $Y(s)$

Part of sol'n driven by IC

“Zero-Input Sol'n”

Part of sol'n driven by input

“Zero-State Sol'n”

Note that $1/(s+a)$ plays a role in both parts...

Hey! $s+a$ is the Characteristic Polynomial!!

Now... the “hard” part is to find the inverse LT of $Y(s)$

Now we apply these general ideas to solving for the output of the previous RC circuit with a unit step input.... $x(t) = u(t)$

$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$$



$$Y(s) = \frac{y(0^-)}{s + 1/RC} + \left[\frac{1/RC}{s + 1/RC} \right] X(s)$$

This “transfers” the input $X(s)$ to the output $Y(s)$
We’ll see this later as “The Transfer Function”

Now... we need the LT of the input...

From the LT table we have:

$$x(t) = u(t) \leftrightarrow X(s) = \frac{1}{s}$$

$$Y(s) = \frac{y(0^-)}{s + 1/RC} + \left[\frac{1/RC}{(s + 1/RC)} \right] \frac{1}{s}$$

Now we have “just a function of s ” to which we apply the ILT...

So now applying the ILT we have:

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{y(0^-)}{s + 1/RC} + \left[\frac{1/RC}{(s + 1/RC)s}\right]\right\}$$

Apply LT to both sides

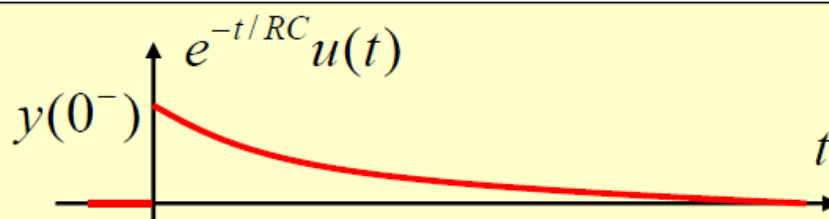
$$y(t) = \underbrace{\mathcal{L}^{-1}\left\{\frac{y(0^-)}{s + 1/RC}\right\}}_{\text{zero-input sol'n}} + \underbrace{\mathcal{L}^{-1}\left\{\left[\frac{1/RC}{(s + 1/RC)s}\right]\right\}}_{\text{zero-state sol'n}}$$

Linearity of LT

This part (zero-input sol'n) is easy...
Just look it up on the LT Table!!

This part (zero-state sol'n) is harder...
It is **NOT** on the LT Table!!

$$\mathcal{L}^{-1}\left\{\frac{y(0^-)}{s + 1/RC}\right\} = y(0^-)e^{-(t/RC)}u(t)$$



So... the part of the sol'n due to the IC (zero-input sol'n) decays down from the IC voltage

Now let's find the other part of the solution... the zero-state sol'n... the part that is driven by the input:

$$y(t) = \mathcal{L}^{-1}\left\{\frac{y(0^-)}{s + 1/RC}\right\} + \mathcal{L}^{-1}\left\{\left[\frac{1/RC}{(s + 1/RC)s}\right]\right\}$$

We can factor this function of s as follows:

$$\mathcal{L}^{-1}\left\{\left[\frac{1/RC}{(s + 1/RC)s}\right]\right\} = \mathcal{L}^{-1}\left\{\left[\frac{1}{s} - \frac{1}{s + 1/RC}\right]\right\}$$

Can do this with
“Partial Fraction
Expansion”, which
is just a “fool-proof”
way to factor

$$= \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s + 1/RC}\right\}$$

Linearity
of LT

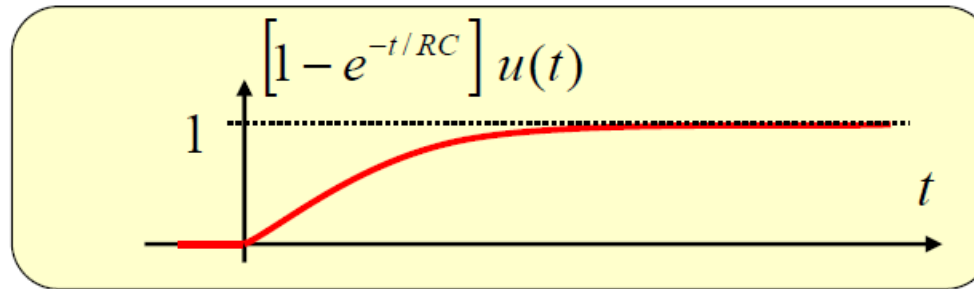
Now... each of these terms
is on the LT table:

$$= u(t)$$

$$= e^{-(t/RC)}u(t)$$

$$= [1 - e^{-(t/RC)}]u(t)$$

So the zero-state response of this system is: $\left[1 - e^{-(t/RC)}\right]u(t)$



Now putting this zero-state response together with the zero-input response we found gives:

$$y(t) = \underbrace{y(0^-)e^{-(t/RC)}u(t)}_{\text{IC Part}} + \underbrace{\left[1 - e^{-(t/RC)}\right]u(t)}_{\text{Input Part}}$$

IC Part

Input Part

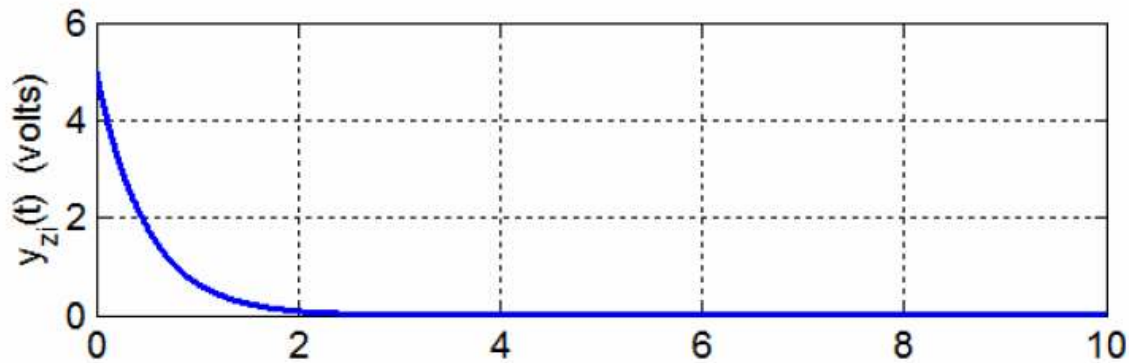
Notice that:

The IC Part “Decays Away”

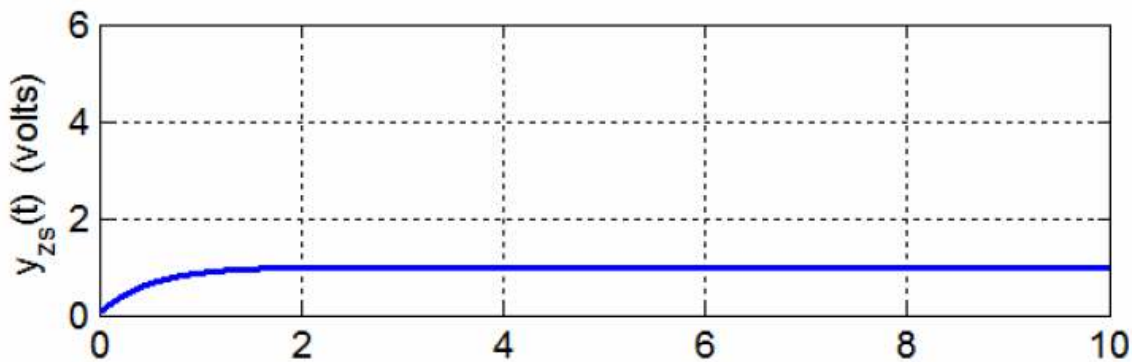
but...

The Input Part “Persists”

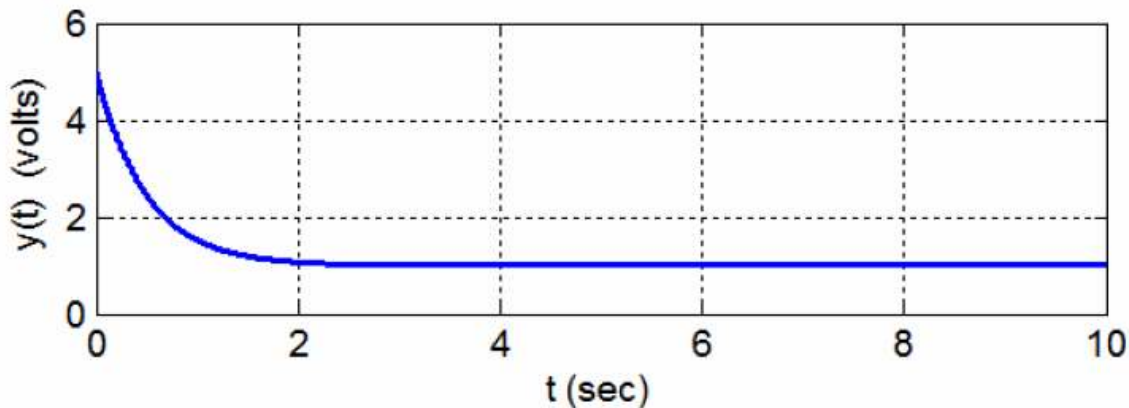
Here is an example for $RC = 0.5$ sec and the initial $V_{IC} = 5$ volts:



**Zero-Input
Response**



**Zero-State
Response**



**Total
Response**

Second-order case

Circuits with two energy-storing devices (C & L, or 2 Cs or 2 Ls) are described by a second-order Differential Equation...

$$\frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

w/ ICs $\dot{y}(0^-)$ & $y(0^-)$

Assume Causal Input

$$x(t) = 0 \quad t < 0$$



$$x(0^-) = 0$$

We solve the 2nd-order case using the same steps:

Take LT of Diff. Equation:

$$\underbrace{\left[s^2 Y(s) - y(0^-)s - \dot{y}(0^-) \right]}_{\text{From 2nd derivative property, accounting for ICs}} + a_1 \underbrace{\left[s Y(s) - y(0^-) \right]}_{\text{From 1st derivative property, accounting for ICs}} + a_0 Y(s) = b_1 \underbrace{s X(s)}_{\text{From 1st derivative property, causal signal}} + b_0 X(s)$$

From 2nd derivative property,
accounting for ICs

From 1st derivative property,
accounting for ICs

From 1st derivative
property, causal signal

Solve for $Y(s)$:

$$Y(s) = \frac{y(0^-)s + \dot{y}(0^-) + a_1 y(0^-)}{s^2 + a_1 s + a_0} + \left[\frac{b_1 s + b_0}{s^2 + a_1 s + a_0} \right] X(s)$$

Part of sol'n driven by IC
“Zero-Input Sol’n”

Note this shows up in both places...
it is the Characteristic Equation

Part of sol'n driven by input
“Zero-State Sol’n”

Note: The role the Characteristic Equation plays here!

It just pops up in the LT method!

The same happened for a 1st-order Diff. Eq...

...and it happens for all orders

Like before...

to get the solution in the time domain find the Inverse LT of $Y(s)$

Nth-Order Case

Diff. eq
of the
system

$$\frac{d^N y(t)}{dt^N} + a_{N-1} \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_M \frac{dx^M(t)}{dt^M} + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

$$\text{For } M \leq N \text{ and } \left. \frac{d^i x(t)}{dt^i} \right|_{t=0^-} = 0 \quad i = 0, 1, 2, \dots, M-1$$

Taking LT and re-arranging gives:

$$Y(s) = \frac{IC(s)}{A(s)} + \frac{B(s)}{A(s)} X(s)$$

LT of the solution (i.e. the LT of the system output)

$$\text{where } \begin{cases} A(s) = s^N + a_{N-1}s^{N-1} + \dots + a_1s + a_0 & \text{“output-side” polynomial} \\ B(s) = b_M s^M + \dots + b_1s + b_0 & \text{“input-side” polynomial} \\ IC(s) = \text{polynomial in } s \text{ that depends on the ICs} \end{cases}$$

Recall: For 2nd order case: $IC(s) = y(0^-)s + [\dot{y}(0^-) + a_1 y(0^-)]$

Consider the case where the LT of $x(t)$ is rational: $X(s) = \frac{N_X(s)}{D_X(s)}$

Then...
$$Y(s) = \frac{IC(s)}{A(s)} + \frac{B(s)}{A(s)} X(s) = \frac{IC(s)}{A(s)} + \frac{B(s)}{A(s)} \frac{N_X(s)}{D_X(s)}$$

This can be expanded like this:
$$Y(s) = \frac{IC(s)}{A(s)} + \frac{E(s)}{A(s)} + \frac{F(s)}{D_X(s)}$$

for some resulting polynomials $E(s)$ and $F(s)$

So... for a system with $H(s) = \frac{B(s)}{A(s)}$ and input with $X(s) = \frac{N_X(s)}{D_X(s)}$
and initial conditions you get:

$$Y(s) = \underbrace{\frac{IC(s)}{A(s)}}_{\text{Zero-Input Response}} + \underbrace{\frac{E(s)}{A(s)}}_{\text{Zero-State Response}} + \underbrace{\frac{F(s)}{D_X(s)}}_{\text{Steady-State Response}}$$

Transient Response **Steady-State Response**

Decays in time domain if
roots of system char. poly.
 $A(s)$ have negative real parts

If all IC's are zero (zero state) $\dot{IC}(s) = 0$

Then:

$$Y(s) = \underbrace{\left[\frac{B(s)}{A(s)} \right]}_{\equiv H(s)} X(s)$$

Called “Transfer Function” of the system...



**Zero-State
Response**

$$Y(s) = \frac{E(s)}{A(s)} + \frac{F(s)}{D_X(s)}$$

**Transient
Response**

**Steady-State
Response**

Summary Comments:

1. From the differential equation one can easily write the $H(s)$ by inspection!
2. The denominator of $H(s)$ is the characteristic equation of the differential equation.
3. The roots of the denominator of $H(s)$ determine the form of the solution...
...recall partial fraction expansions

BIG PICTURE: The roots of the characteristic equation drive the nature of the system response... we can now see that via the LT.

We now see that there are three contributions to a system's response:

zero-input
resp.

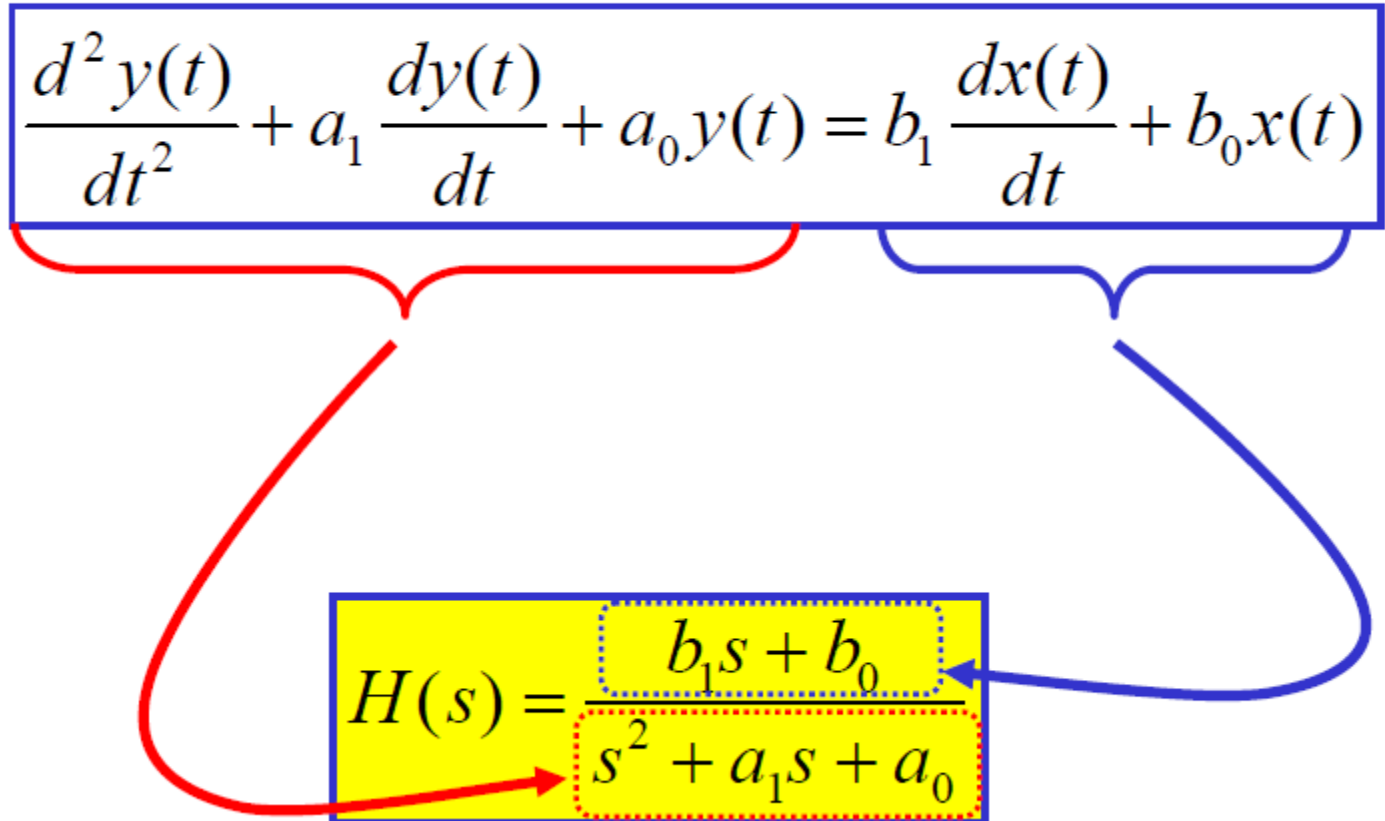
1. The part driven by the ICs
 - a. This will decay away if the Ch. Eq. roots have negative real parts

zero-state
resp.

2. A part driven by the input that will decay away if the Ch. Eq. roots have negative real parts ... “Transient Response”
3. A part driven by the input that will persist while the input persists... “Steady State Response”

Poles and Zeros

it is possible to directly identify the TF $H(s)$ from the Diff. Eq.:


$$\frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_1 \frac{dx(t)}{dt} + b_0 x(t)$$
$$H(s) = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}$$

Poles and Zeros

So... we know that $H(s)$ is completely described by the Diff. Eq.... Therefore we should expect that we can tell a lot about a system by looking at the structure of the transfer function $H(s)$... This structure is captured in the idea of “Poles” and “Zeros”...

Poles and Zeros of a system

Given a system with Transfer Function:

$$H(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_1 s + b_0}{s^N + a_{N-1} s^{N-1} + \dots + a_1 s + a_0}$$

$\leftarrow B(s)$
 $\leftarrow A(s)$

We can factor $B(s)$ and $A(s)$: (Recall: $A(s)$ = characteristic polynomial)

$$H(s) = \frac{b_M (s - z_1)(s - z_2) \dots (s - z_M)}{(s - p_1)(s - p_2) \dots (s - p_N)}$$

Assume any common factors in $B(s)$ and $A(s)$ have been cancelled out

Note: $H(s)|_{s=z_i} = 0$ $i = 1, 2, \dots, M$

$\{z_i\}$ are called "zeros of $H(s)$ "

$H(s)|_{s=p_i} = \infty$ $i = 1, 2, \dots, N$

$\{p_i\}$ are called "poles of $H(s)$ "

Note: p_i are the roots of the char. polynomial

Poles and Zeros

Note that knowing the sets $\{z_i\}_{i=1}^M$ & $\{p_i\}_{i=1}^N$

tells us what $H(s)$ is: (up to the multiplicative scale factor b_M)

$-b_M$ is like a gain (i.e. amplification)

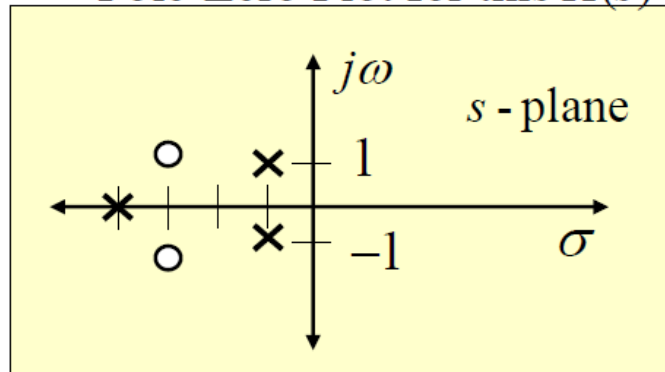
Pole-Zero Plot

This gives us a graphical view of the system's behavior

Example:
$$H(s) = \frac{2s^2 + 12s + 20}{s^3 + 6s^2 + 10s + 8} = \frac{2(s + 3 - j)(s + 3 + j)}{(s + 4)(s + 1 - j)(s + 1 + j)}$$

Real coefficients $\Rightarrow$ complex conjugate pairs

Pole-Zero Plot for this $H(s)$



x denotes a pole

o denotes a zero

From the Pole-Zero Plots we can Visualize the TF function on the s-plane:

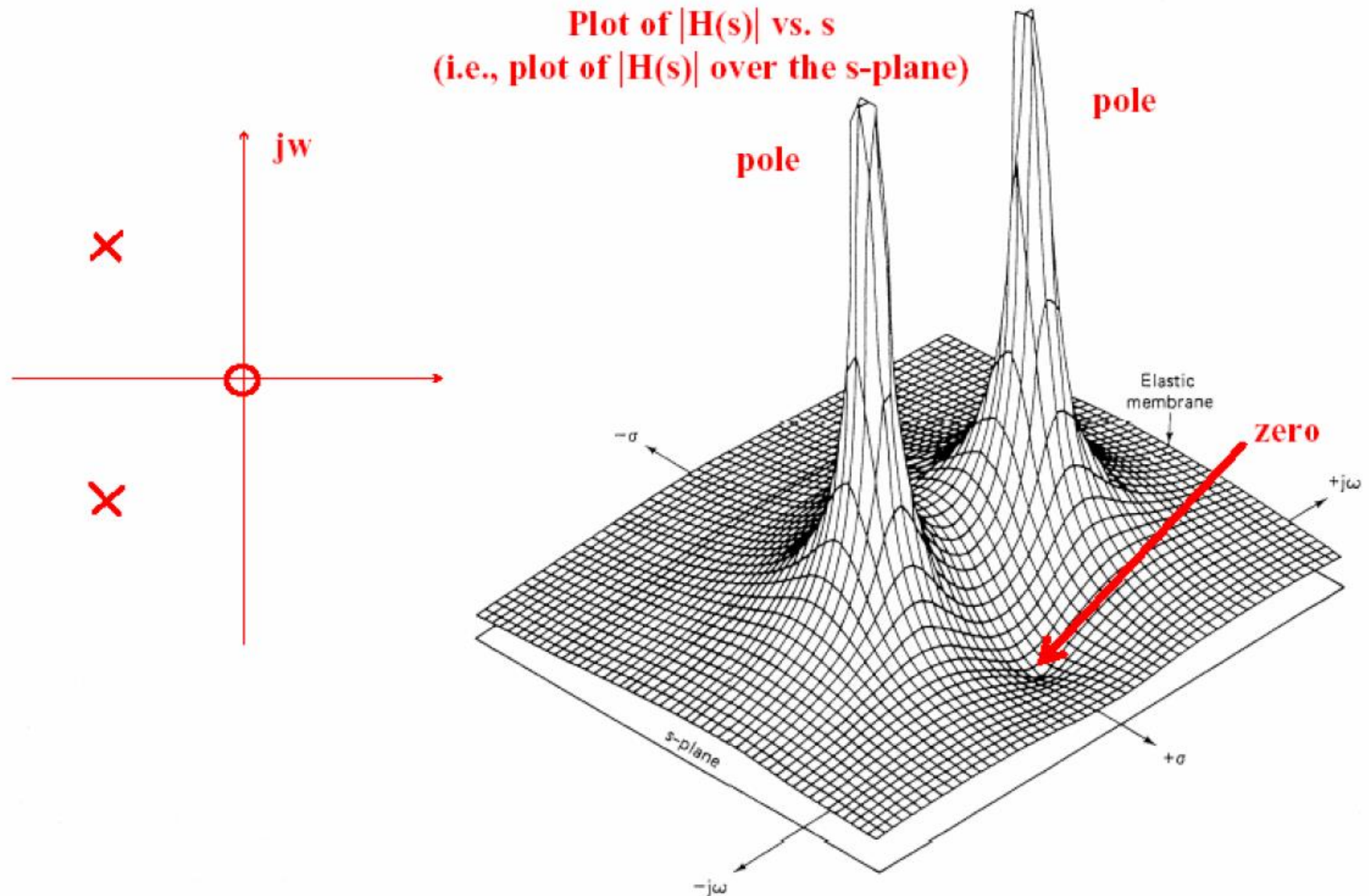
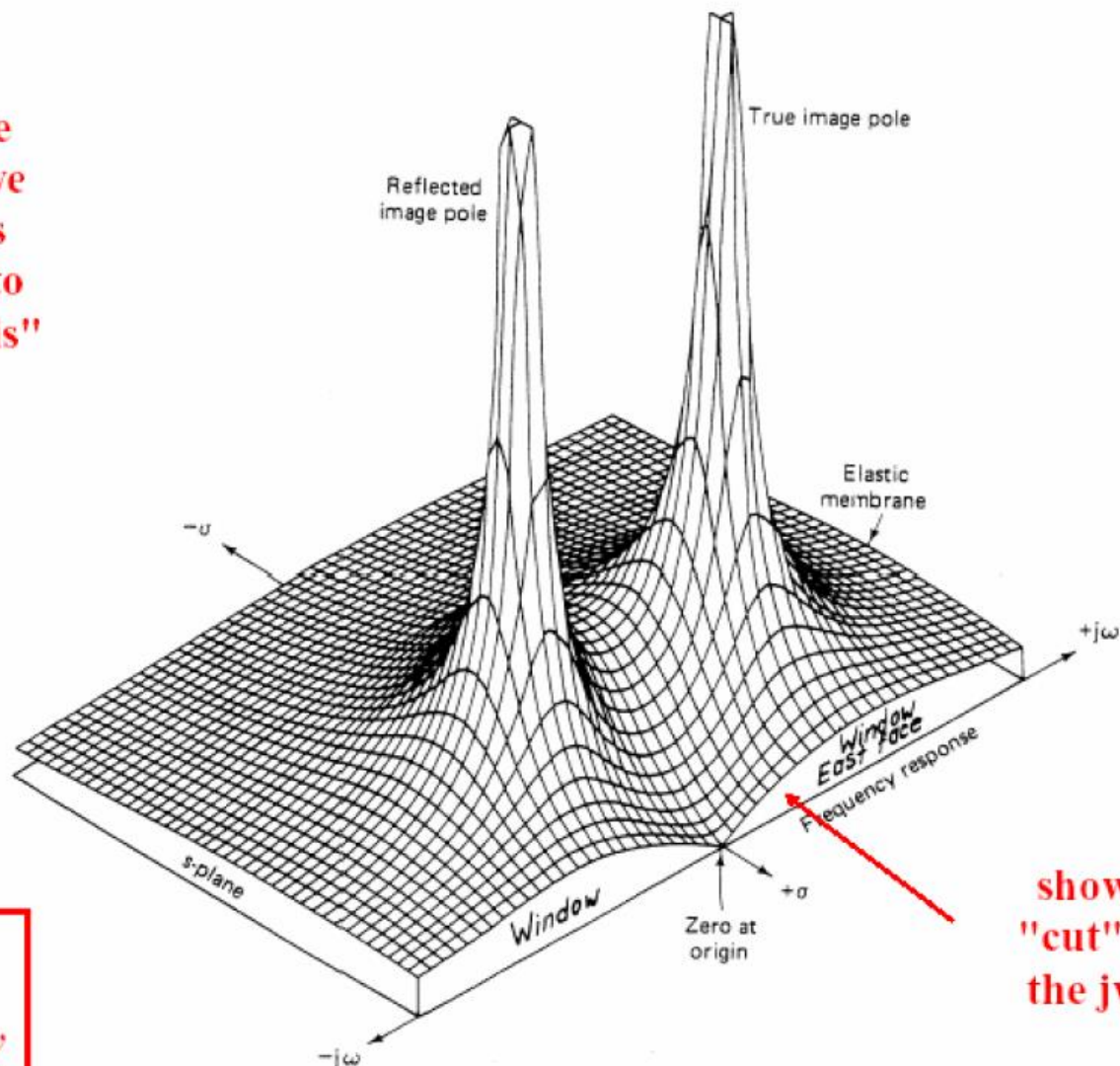


Fig. 9-4. Shape of elastic membrane for a pair of poles and a zero.

From our Visualization of the TF function on the s-plane we can see the Freq. Resp.:

To get the frequency response $H(\omega)$ from the transfer function $H(s)$ we replace s by $j\omega$... this is graphically equivalent to "cutting along the $j\omega$ axis"

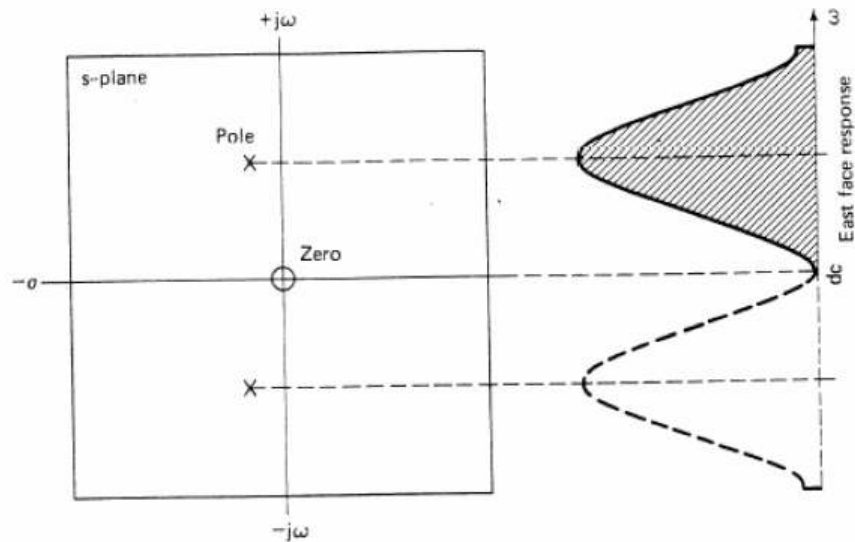


shows the "cut" along the $j\omega$ axis

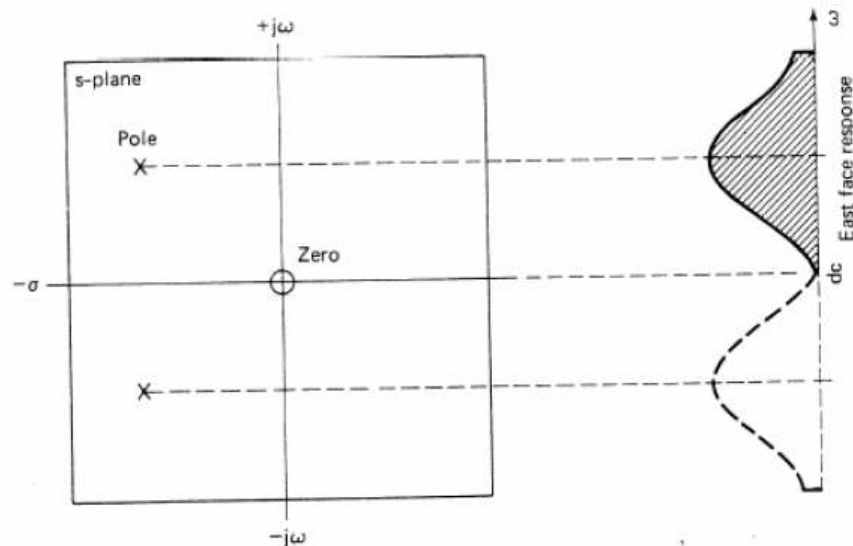
Plots from my favorite book on Op Amps: "Operational Amplifier: Characteristics and Applications" by Robert G. Irvine, Prentice-Hall, 1981

Fig. 9-5. Pole-zero diagram showing the east face.

Can also look at a pole-zero plot and see the effects on Freq. Resp.



(a) Frequency response for pole close to $j\omega$ axis



(b) Frequency response for pole far from $j\omega$ axis

Fig. 9-6. Frequency response versus pole location.

As the pole moves closer to the $j\omega$ axis it has a stronger effect on the frequency response $H(\omega)$. Poles close to the $j\omega$ axis will yield sharper and taller bumps in the frequency response.

By being able to visualize what $|H(s)|$ will look like based on where the poles and zeros are, an engineer gains the ability to know what kind of transfer function is needed to achieve a desired frequency response... then through accumulated knowledge of electronic circuits (requires experience accumulated AFTER graduation) the engineer can devise a circuit that will achieve the desired effect.

Laplace Transform Table

Time Signal	Laplace Transform		
$u(t)$	$1/s$		
$u(t) - u(t - c), \quad c > 0$	$(1 - e^{-cs})/s, \quad c > 0$		
$t^N u(t), \quad N = 1, 2, 3, \dots$	$\frac{N!}{s^{N+1}}, \quad N = 1, 2, 3, \dots$		
$\delta(t)$	1		
$\delta(t - c), \quad c \text{ real}$	$e^{-cs}, \quad c \text{ real}$		
$e^{-bt} u(t), \quad b \text{ real or complex}$	$\frac{1}{s + b}, \quad b \text{ real or complex}$		
$t^N e^{-bt} u(t), \quad N = 1, 2, 3, \dots$	$\frac{N!}{(s + b)^{N+1}}, \quad N = 1, 2, 3, \dots$		
$\cos(\omega_o t) u(t)$	$\frac{s}{s^2 + \omega_o^2}$	$t \sin(\omega_o t) u(t)$	$\frac{2\omega_o s}{(s^2 + \omega_o^2)^2}$
$\sin(\omega_o t) u(t)$	$\frac{\omega_o}{s^2 + \omega_o^2}$	$A e^{-\zeta \omega_n t} \sin\left[\left(\omega_n \sqrt{1 - \zeta^2}\right) t\right] u(t)$	$\frac{\alpha}{s^2 + 2\zeta \omega_n s + \omega_n^2}$
$\cos^2(\omega_o t) u(t)$	$\frac{s^2 + 2\omega_o^2}{s(s^2 + 4\omega_o^2)}$	where: $A = \frac{\alpha}{\omega_n \sqrt{1 - \zeta^2}}$	
$\sin^2(\omega_o t) u(t)$	$\frac{2\omega_o^2}{s(s^2 + 4\omega_o^2)}$	$A e^{-\zeta \omega_n t} \sin\left[\left(\omega_n \sqrt{1 - \zeta^2}\right) t + \phi\right] u(t)$	$\beta \frac{s + \alpha}{s^2 + 2\zeta \omega_n s + \omega_n^2}$
$e^{-bt} \cos(\omega_o t) u(t)$	$\frac{s + b}{(s + b)^2 + \omega_o^2}$	$A = \beta \sqrt{\frac{(\alpha - \zeta \omega_n)^2}{\omega_n^2 (1 - \zeta^2)} + 1} \quad \phi = \tan^{-1}\left(\frac{\omega_n \sqrt{1 - \zeta^2}}{\alpha - \zeta \omega_n}\right)$	
$e^{-bt} \sin(\omega_o t) u(t)$	$\frac{\omega_o}{(s + b)^2 + \omega_o^2}$	$t e^{-bt} \cos(\omega_o t) u(t)$	$\frac{(s + b)^2 - \omega_o^2}{((s + b)^2 + \omega_o^2)^2}$
$t \cos(\omega_o t) u(t)$	$\frac{s^2 - \omega_o^2}{(s^2 + \omega_o^2)^2}$	$t e^{-bt} \sin(\omega_o t) u(t)$	$\frac{2\omega_o (s + b)}{((s + b)^2 + \omega_o^2)^2}$

Laplace Transform Properties

Property Name	Property	
Linearity	$ax(t) + bv(t)$	$aX(s) + bV(s)$
Right Time Shift (Causal Signal)	$x(t - c), \quad c > 0$	$e^{-cs} X(s)$
Time Scaling	$x(at), \quad a > 0$	$\frac{1}{a} X(s/a), \quad a > 0$
Multiply by t^n	$t^n x(t), \quad n = 1, 2, 3, \dots$	$(-1)^n \frac{d^n}{ds^n} X(s), \quad n = 1, 2, 3, \dots$
Multiply by Exponential	$e^{at} x(t), \quad a \text{ real or complex}$	$X(s - a), \quad a \text{ real or complex}$
Multiply by Sine	$\sin(\omega_o t) x(t)$	$\frac{j}{2} [X(s + j\omega_o) - X(s - j\omega_o)]$
Multiply by Cosine	$\cos(\omega_o t) x(t)$	$\frac{1}{2} [X(s + j\omega_o) + X(s - j\omega_o)]$
Time Differentiation 2 nd Derivative n^{th} Derivative	$\dot{x}(t)$ $\ddot{x}(t)$ $x^{(N)}(t)$	$sX(s) - x(0)$ $s^2 X(s) - sx(0) - \dot{x}(0)$ $s^N X(s) - s^{N-1} x(0) - s^{N-2} \dot{x}(0) - \dots - sx^{(N-2)}(0) - x^{(N-1)}(0)$
Time Integration	$\int_{-\infty}^t x(\lambda) d\lambda$	$\frac{1}{s} X(s)$
Convolution in Time	$x(t) * h(t)$	$X(s)H(s)$
Initial-Value Theorem	$x(0) = \lim_{s \rightarrow \infty} [sX(s)]$ $\dot{x}(0) = \lim_{s \rightarrow \infty} [s^2 X(s) - sx(0)]$ $x^{(N)}(0) = \lim_{s \rightarrow \infty} [s^{N+1} X(s) - s^N x(0) - s^{N-1} \dot{x}(0) - \dots - sx^{(N-1)}(0)]$	
Final-Value Theorem	If $\lim_{t \rightarrow \infty} x(t)$ exists, then $\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s)$	